

1.9.3

Bhoomika L - EE25BTECH11014

August 2025

Question

$AOBC$ is a rectangle whose three vertices are $(0, -3)$ $(0, 0)$ $(4, 0)$. The length of its diagonal is_____

Theoretical Solution

Given the points **A**, **O** and **B** :

Point	vector
Point A	$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$
Point O	$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$
Point B	$\begin{pmatrix} 0 \\ -3 \end{pmatrix}$

Table: Position Vectors of the points on rectangle.

Theoretical Solution

Determining the Coordinates of Point C:

$$\overrightarrow{OA} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \quad \overrightarrow{OB} = \begin{pmatrix} 0 \\ -3 \end{pmatrix} \quad (1)$$

Since **C** is opposite to **O** in the rectangle,

$$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{OB} \quad (2)$$

$$\Rightarrow \begin{pmatrix} 4 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad (3)$$

$$\therefore \mathbf{C} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad (4)$$

Theoretical solution

We know that the length of the diagonal vector is magnitude of the vector $\vec{OC}$.

Let $\mathbf{P} = \vec{OC}$

$$\mathbf{P} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad (5)$$

$$|\mathbf{P}| = \sqrt{\mathbf{P}^T \cdot \mathbf{P}} \quad (6)$$

Theoretical Solution

$$\mathbf{P}^T \cdot \mathbf{P} = \begin{pmatrix} 4 & -3 \end{pmatrix} \begin{pmatrix} 4 \\ -3 \end{pmatrix} = 4^2 + (-3)^2 = 16 + 9 = 25 \quad (7)$$

$$|\mathbf{P}| = \sqrt{25} = 5 \quad (8)$$

Therefore the length of the diagonal is 5.

```
#include <stdio.h>
#include <math.h>

int main() {
    // Coordinates of points
    int x1 = 0, y1 = -3; // A
    int x2 = 4, y2 = 0; // B

    // Calculate diagonal length using distance formula
    double diagonal = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));

    printf(The length of the diagonal of rectangle AOB is: %.2f\n, diagonal);

    return 0;
}
```


Python Code

```
import numpy as np
import matplotlib.pyplot as plt
import ctypes

# Function to compute the distance between two points using the
  distance formula
def distance(point1, point2):
    return np.sqrt((point2[0] - point1[0])**2 + (point2[1] -
        point1[1])**2)

# Define the coordinates of points A, B, and O
A = np.array([0.0, -3.0])
O = np.array([0.0, 0.0])
B = np.array([4.0, 0.0])
```

```
# Calculate the coordinates of C (since A0BC is a rectangle)
C = np.array([4.0, -3.0])

# Calculate the length of the diagonal AC
diagonal_length = distance(A, C)

# Print the length of the diagonal
print(fThe length of the diagonal AC is {diagonal_length:.2f}
      units.)
```

```
# Now, plot the rectangle and the diagonal
# Generate the rectangle's lines for plotting
rectangle_x = [A[0], O[0], B[0], C[0], A[0]]
rectangle_y = [A[1], O[1], B[1], C[1], A[1]]

plt.figure(figsize=(6, 6))
plt.plot(rectangle_x, rectangle_y, label='Rectangle AOBC', color=
        'blue')

# Plot points A, B, O, and C
points = np.array([A, O, B, C])
plt.scatter(points[:, 0], points[:, 1], color='red')
```

```
# Annotate points A, B, O, and C
point_labels = [f'A {tuple(A)}', f'O {tuple(O)}', f'B {tuple(B)}',
                f'C {tuple(C)}']
for i, txt in enumerate(point_labels):
    plt.annotate(txt,
                 (points[i, 0], points[i, 1]),
                 textcoords=offset points,
                 xytext=(10, 5),
                 ha='center')
```

```
# Set plot details
plt.xlabel('$x$')
plt.ylabel('$y$')
plt.title(f'Rectangle AOB with diagonal AC = {diagonal_length:.2f}')
plt.grid(True)
plt.axis('equal')
plt.legend(loc='best')

# Save and show the plot
plt.savefig('../Figs/fig2.png')
plt.show()
```

Plot

